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An analysis of rock magnetic data

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The data obtained from numerous palaeomagnetic measurements made during the past decade have shown that while the geologically younger rock formations are magnetized in directions close to that of the present earth's magnetic field, the remanent magnetic polarizations of older rocks depart markedly from this pattern. These observations are widely held by many workers to suggest that the main continental land masses have undergone movements relative to one another during the past. The present paper gives an account of a new analysis of the available data, making a minimum number of theoretical assumptions about the ways in which the rocks became magnetized and about the origin of the geomagnetic field.

The results of this analysis strongly support the supposition that the observed wide divergence between the directions of the remanent magnetic vectors of older rocks and that of the present earth's field is systematic, and not a result of purely random processes occurring throughout geological time. The most reasonable explanations of the phenomenon appear to be that (*a*) the directions of magnetization of the earlier rocks have been changed by some widespread physical or geological processes since the time of their formation, (*b*) the earth's magnetic field has had strong multipolar components in past geological ages, (*c*) a relative drift of the continents across the earth's mantle has occurred. Of these hypotheses, (*c*) appears to be the most plausible. On the tentative assumption that the rock magnetic results can be explained by continental drift, it is possible to estimate the ancient latitude and the orientation relative to the earth's rotational axis, of each continent, although by palaeomagnetic measurements alone changes in relative longitude cannot be revealed.

1. Introduction

In recent years many workers* have deduced from measurements of the direction of remanent magnetism in rocks that the main continental land masses have moved markedly relative to the earth's rotational axis and also relative to each other. Others* have held that the palaeomagnetic evidence for such an important conclusion is not strong, and that there may be other plausible explanations of the observed facts of rock magnetism. There is also a wide divergence of opinion among authoritative geologists on the question of past relative movements of the continents. The postulates of those who advocate drift in some form or another, such as Wegener (1941) and du Toit (1937), have been severely critized on geological grounds by others, such as, for instance, Umbgrove (1951), Stille (1936) and Bucher (1951), and on theoretical grounds by Jeffreys (1959). A summary of the present position has been given by Kuenen (1950), who concludes that much of the evidence is indecisive and that on the balance most geologists appear to have lost faith in continental drift as a working hypothesis.

This paper deals only with the rock magnetic evidence pertaining to the problem. Recently a comprehensive survey of the ancient pole positions calculated from palaeomagnetic measurements has been made by Irving† (1959), and the results

* Papers on the palaeomagnetic evidence for and against the hypothesis of continental drift are too numerous to list here in detail. The majority, including the more important ones, appear in the References at the end of this paper.

† Previous surveys have also been made by a number of others prominent among whom are Runcorn (1956*a*, *b*), Creer *et al.* (1954, 1957, 1958), and Du Bois (1957).

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have been used to give strong support to the hypothesis of continental drift. Following customary practice he calculates the magnetic pole position corresponding to each set of measurements. It has been known for some years that, in recent geological times, the poles derived by palaeomagnetic measurements for different continents lie close to the present geographical pole, whereas wide differences of pole position occur in earlier geological times. These are essentially the results which have been widely held by rock magnetists to give strong evidence that the continents must have drifted relative to each other during geological history.

To inquire still further into the possibility that some other explanation of the observed experimental data might be found, we have made a new systematic analysis of the direction of the magnetization of the rock specimens in a somewhat different and supplementary way to that used by Irving. We have used essentially the same material, but have added a few recent results not previously available. All information from five continents for which sufficient numerical data have been published to date is included. No analysis will be given here of the intensities of the magnetization, from which it is hoped ultimately to obtain an estimate of the magnitude of the ancient field, nor of the phenomena of magnetic reversals. As regards the former there is no conclusive evidence pointing to either a larger or smaller field in ancient times; a reasonable guess is that it has not changed either way by as much as a factor of two or three. Except in a few cases of ambiguity, the analysis is independent of whether the reversed rocks are due to the reversal of the earth's field or to some physical or chemical mechanism.

The main analysis of the observations will be based on a minimum of theoretical assumptions, both about the various ways in which the rocks acquire their magnetization and about the still very hypothetical mechanisms in the earth which give rise to the terrestrial magnetic field.

A major object of the analysis is to assist the planning of future rock collecting and measurement programmes, by drawing attention to those land areas, types and ages of rocks where more measurements are urgently needed.

2. Collected information in sites and measurements

The available data are tabulated in tables 1 to 5 of appendix II for the following land masses: Western Europe, North America, Australia, India and Africa. The dates T assigned to the rocks are those of Holmes B time scale (Holmes 1947); see appendix III. No rocks older than 500 My (million years) are included. Although more recent age determinations, from improved radioactive methods, have lately led to the proposal of a rather longer time scale (Mayne, Lambert & York 1959), that of Holmes has been retained pending the general acceptance of a new index of absolute age. In estimating the dates of formation of rocks the majority of workers in palaeomagnetism refer merely to their geological periods. For the purpose of this analysis we have arbitrarily taken the numerical age T as that assigned to the middle of the period on the chosen scale.

The basic quantities determined by palaeomagnetic measurements on a suite of rocks from a given locality are the mean declination D, and the mean inclination I,

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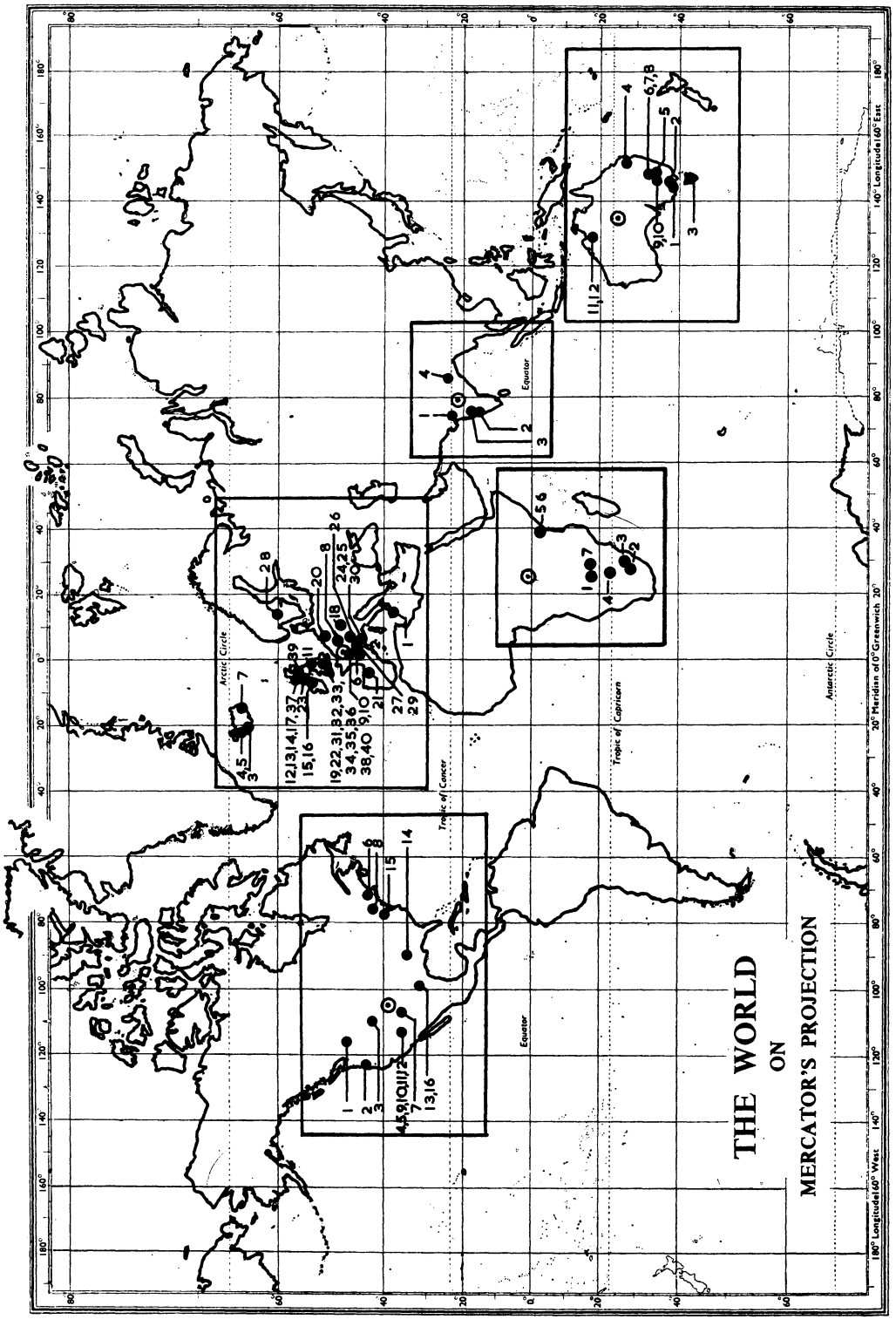


FIGURE 1. Distribution of sampling localities. (Base map by courtesy of George Philip and Son, Ltd.)

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which together define the remanent magnetic vector. The measured inclination I has certainly in most cases, and presumably in all, been corrected for the tilt of the geological formation. It is taken as positive when the remanent magnetic vector is downwards as is the direction of the present earth's field in the northern hemisphere. In addition to the pole positions tabulated by Irving, we have included the actual measurements of D and I in the tables, together with the geographical latitude λ_0 and the longitude L_0 of each site, and also some additional derived quantities. The map in figure 1 shows the various sampling localities in each of the five continents. One object of this compilation of data is to facilitate their independent analysis by other workers who wish to avoid the necessity of collecting the data from some sixty publications in a dozen different journals. To this end all the basic information and measurements are collected in columns I to VIII, and all the derived data in columns IX to XVI of the tables.

The methods of analysis are chosen so as to yield, wherever possible, results expressible in numerical form, so that quantitative comparisons with various hypothetical explanations are made possible. Moreover, they are such as to make easy the incorporation of new data as they become available, so as to give more accurate results.

3. Analysis of data

3·1. *The values of I and D*

In columns III and VII of tables 1 to 5 are listed the geological age T and the measured values of D and I for the various individual sites in the continents concerned. It is desirable to display these data with the minimum of assumptions about the eventual explanation of the results. To this end it is useful, in making a preliminary analysis, to compare the direction of the remanent magnetic vector of the rocks with that of the recent earth's field at the locality concerned. It is generally accepted that the recent geomagnetic field, averaged over periods of the order of a few thousand years, has coincided closely with that which would be expected of a magnetic dipole situated at the centre of the earth and directed along the rotational axis. The field due to such a dipole would have zero declination, and its inclination I_0 would be given by

$$\tan I_0 = 2 \tan \lambda_0. \tag{1}$$

It is convenient to substitute for the measured inclination I the difference ΔI between this quantity and the inclination of such an axial dipole field. This parameter is defined in the following way

$$\left.\begin{aligned} &(1)\ \text{For normally magnetized rocks*} \\ &\qquad \Delta I = I - I_0. \\ &(2)\ \text{For reversely magnetized rocks*} \\ &\qquad \Delta I = -I - I_0. \end{aligned}\right\} \tag{2}$$

* The criterion by which a rock will be classed as normal or reversed in magnetic polarization is given in appendix I.

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Furthermore, following usual practice, the values of D listed in column III of the tables are given in degrees measured clockwise from geographical north, and, for reasons which will be stated later, we will substitute a quantity ψ which we will term the rotation. This will also be measured from north, but will be defined when normal as positive when in the eastern sector and negative when in the western sector thus

$$\left.\begin{aligned} &\text{if} \quad 0° < D < 180°, \quad \psi = D \text{ or } D-180°;* \\ &\text{if } 180° < D < 360°, \quad \psi = D-360° \text{ or } D-180°.* \end{aligned}\right\} \tag{3}$$

The advantage of employing these two quantities ΔI and ψ rather than I and D is that they give a more direct numerical estimate of the significance of the results. Thus if (a) the geomagnetic field had always been that of an axial dipole, (b) the directions of magnetization of the rocks faithfully represented that of the magnetizing field, and (c) the continental land masses had always occupied the same positions relative to the earth's rotational axis, then the values of both ΔI and ψ obtained by measurements on rocks from all lands would be zero for all values of T.

In figures 2 to 5, (a) and (b), are shown the values of ΔI and ψ plotted against T for the four continents of Europe, North America, Australia, and India, for which there are adequate data. For the continents of Asia, Africa and South America the information is as yet too meagre to permit analysis. The dashed curves in figures 2(a), etc., show the mean value of I, and represent the full curve giving ΔI, displaced upwards by the angle I_0. Though there is in some cases a wide scatter of the points, it is clear that though the mean values of both quantities ΔI and ψ are sensibly zero in recent geological times, they depart widely from zero during most of ancient geological time. For instance, no amount of misdating could explain the consistent deviation of ΔI or ψ from zero for Europe, North America or India, figures 2(a), 3(a) and 5(a), though extreme misdating might do so for Australia in the case of ΔI but not for ψ. We conclude that the mean curves drawn by eye through the points of figures 2 to 5, (a) and (b), do have some real physical significance. The large scatter of points about the mean curves may be due, in addition to misdating, to inadequate sampling, large deviation of the earth's field from that of an axial dipole, physico-chemical processes in the rocks and local geological movements.

By comparing the values of ΔI and ψ in figures 2 to 5, (a) and (b), for each of the four continents, it is seen that they are closely correlated: in general, large values of ΔI are found along with large values of ψ. This reinforces the view that the deviation of the rock magnetic directions from that of the present axial dipole must have some physical significance and cannot be due only to some effect giving a random scatter. This correlation can be seen in another way in figure 7. (See also §3·3.)

* Again, as for ΔI, because of the occurrence of rock-magnetic reversals there is an inherent ambiguity of 180° in the value of ψ. The method adopted for resolving this ambiguity is given in appendix I.

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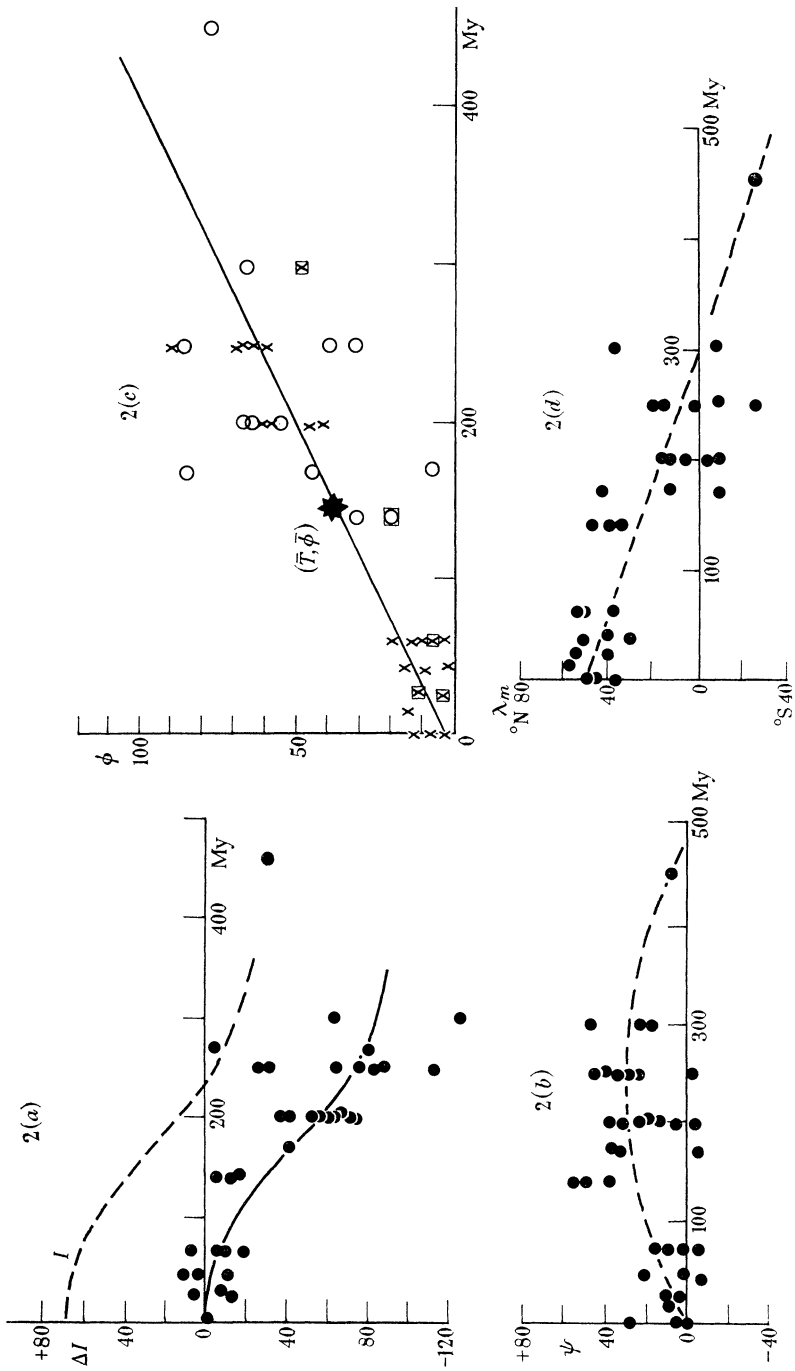


FIGURE 2. Europe. Key to figures 2, 3, 4 and 5, (c): ×, igneous; ○, sedimentary; □, N + R.

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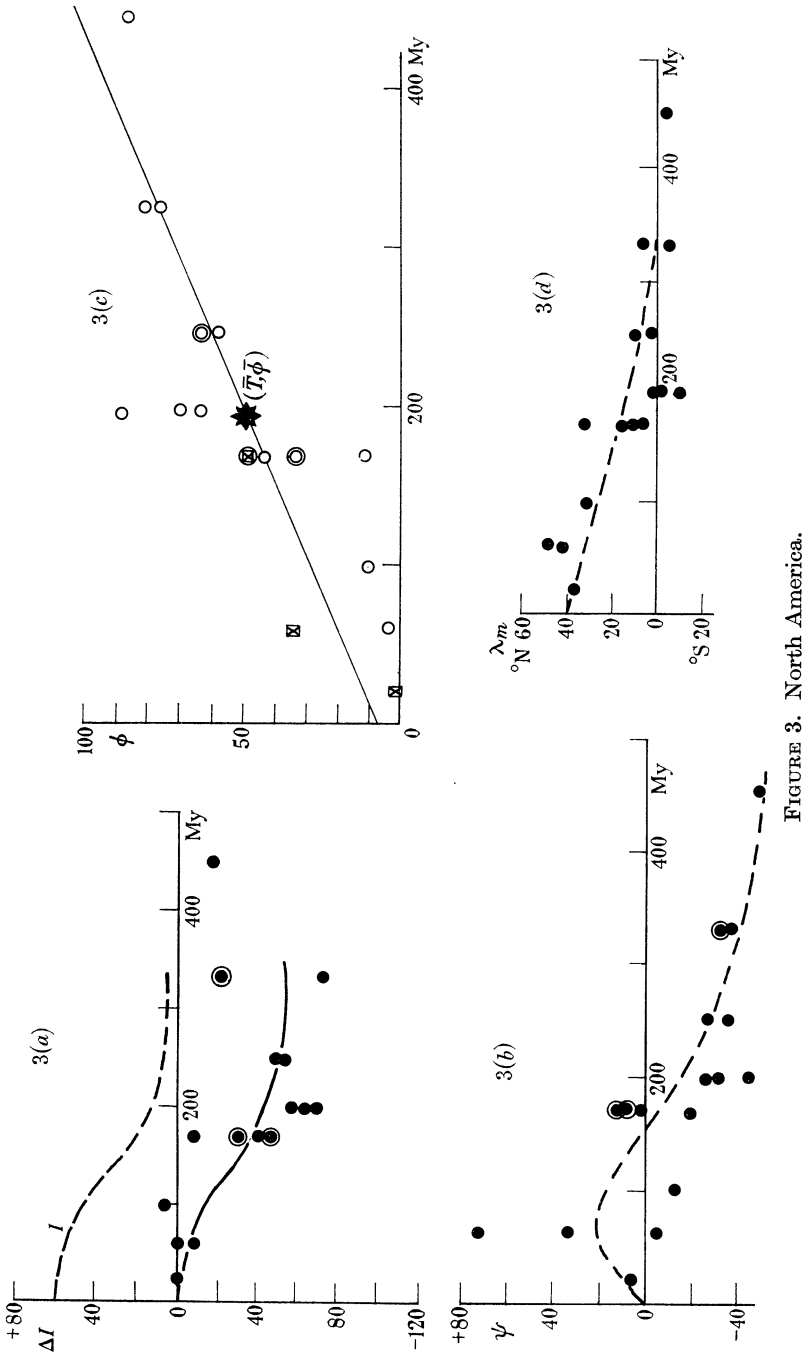


FIGURE 3. North America.

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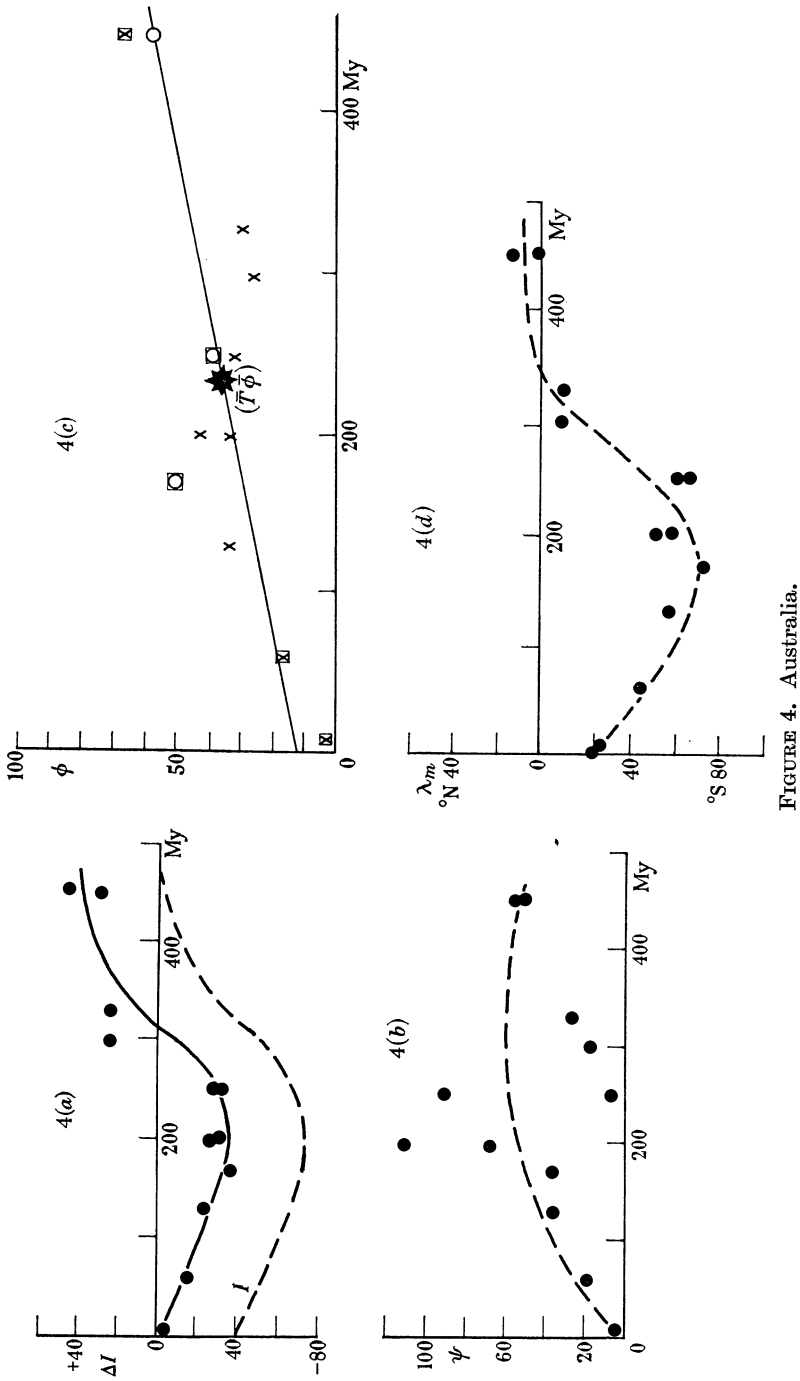


FIGURE 4. Australia.

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3·2. *The use of the divergence* ϕ

It is also found to be useful particularly for the discussion in § 4·1, to combine the two quantities ψ and I into a single angle ϕ (see figure 6) which we call the divergence. This is the angle between the direction of the remanent vector of the rocks at any site and that of the local axial dipole field, and is given by

$$\cos\phi = \cos\psi\cos I\cos I_0 + \sin I\sin I_0. \tag{4}$$

As with ψ and ΔI, ϕ is zero in the present epoch, and would always have been zero if the three assumptions (*a*), (*b*) and (*c*) given above were satisfied. The values of ϕ plotted against T are given in figures 2 to 5, (*c*).

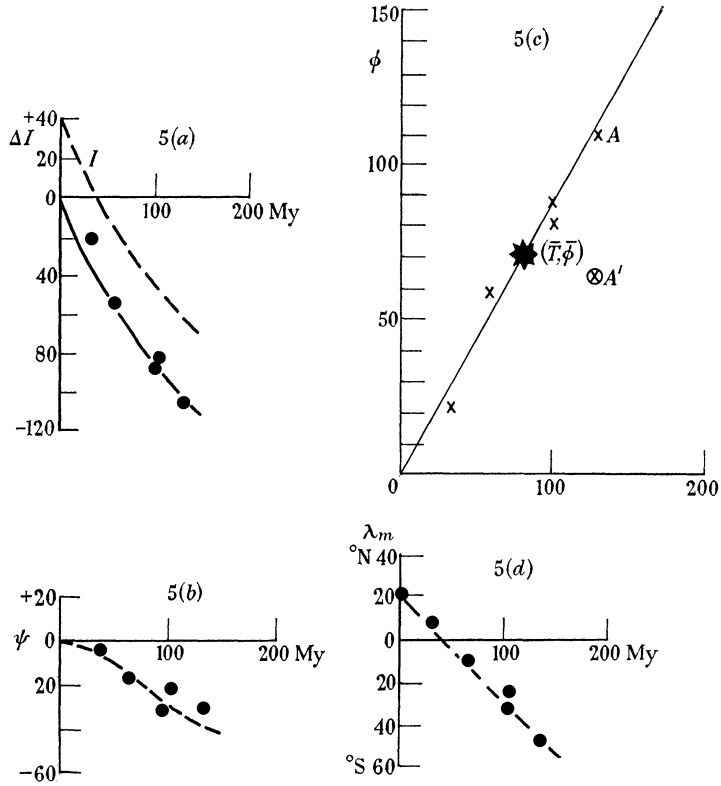


FIGURE 5. India. (A', alternative point for A, being its mirror image through $\phi = 90°$.)

The use of the single variable ϕ rather than ΔI and ψ is exactly equivalent to using, in plane geometry, the arithmetic value of a radius vector rather than its two Cartesian components x and y. Some information is thrown away but the simplicity of having one parameter only to consider for each site instead of two is a considerable gain. In particular, as will be seen below, this method of presentation facilitates the numerical analysis of the results.

Of course, a similar use of ΔI and ψ is possible and would, in principle, be preferable when more and better data are available. A disadvantage of using ψ alone is that, owing to errors of measurement of the vector direction, it becomes

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almost indeterminate when I is near $\pm 90°$. This occurs for Australia between 150 and 250 My ago, and results in the large observed scatter of ψ in figure 4(b).

The widespread phenomenon of reversely magnetized rocks sometimes introduces a possible ambiguity in ΔI and ψ and so in the numerical value of ϕ, as given by (4). For reasons explained in appendix I, in constructing figures 2 to 5, (c), the choice between the two possible values of ϕ is made so as to avoid large abrupt changes in $\mathrm{d}\phi/\mathrm{d}T$. The corresponding values of ΔI and ψ are then selected. The only point in the figures appreciably affected by the above ambiguity is that for India at $T = 150$ My. An alternative possible position for this point is also shown in figure 5(c).

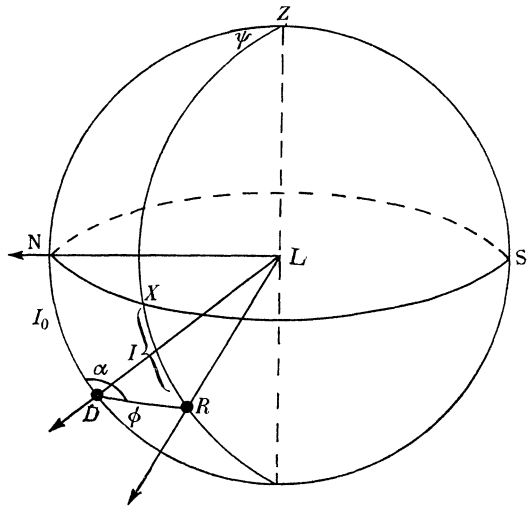


FIGURE 6. Derivation of ϕ and α.

Despite all the uncertainties it is apparent from figures 2 to 5, (c), that ϕ increases with T from values close to zero at $T = 0$ to values sometimes of the order of 90° or more at $T = 5 \times 10^8$ y. The most rapid variation occurs in India where ϕ rises from zero today to more than 90° about 100 My ago.*

3·3. *Significance of the increase of ϕ with T*

Neither the steady increase of divergence with increasing geological age, nor the paucity of low values of ϕ for large values of T are in themselves necessarily physically significant. Indeed if we ignore all other available data it is possible to explain the observed distribution of points in figures 2 to 5, (c), as a manifestation of purely random processes, which have tended to change the original magnetic polarizations arbitrarily throughout geological time. However, if the phenomenon

* Owing to the method adopted for assigning the age T there may well be errors in the positions of the five points shown in figure 5 for India, since there is some doubt as to the exact geological periods of the formations concerned. It may be fortuitous that these points apparently lie on smooth curves on the respective diagrams.

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were indeed due to a sequence of fortuitous events changing the directions of remanent magnetism in this way, one would expect the magnetic vectors to have conical symmetry about the present dipole field.

In order to examine whether this is so we have to define a new angle α as indicated in figure 6, where $SLXN$ represents the horizontal plane at the locality L, LD is the direction of the local dipole field with inclination I_0, and LR is the direction of remanent magnetization of a particular rock having inclination I and rotation ψ. The angle α, which we will term the deviation, is given by

$$\sin\alpha = \frac{\cos I \sin\psi}{\sin\phi}. \tag{5}$$

In figures 7 (*a*) to (*d*) are shown plots of α against ϕ for each of the four continents, with values of α ranging from 0 to 360°. It is evident from these diagrams that the configuration of points is by no means haphazard. For the continents of Europe and India in particular the range of α is less than 90° and for North America and Australia the distribution, although more widely dispersed, still appears to be incompatible with the assumption of purely fortuitous scatter. Some differentiation in age has been made by plotting the vectors for Palaeozoic (200 to 500 My), Mesozoic (100 to 200 My), and Cenozoic (0 to 100 My) eras with different notation. For Europe in particular (figure 7 (*a*)), for which the data are most comprehensive, the general increase of ϕ with age is unaccompanied by any marked increase in the range of α, and there is, moreover, obvious correlation between the two quantities. It is the acceptance of this evidence of a systematic rather than random change of ϕ and α with T that makes the statistical analysis in § 3·4 appropriate. It is necessary here to introduce a word of warning. The rocks suitable for palaeomagnetic measurement are limited to a very few lithological types, and of these many are either very weakly or unstably magnetized or give inconsistent results, so that it is often difficult to establish a criterion for the acceptance or rejection of a series of experimental measurements. It inevitably follows that when an experimenter is confronted with a set of data he is liable to a bias towards rejecting those which fit least well into the already established pattern. The empirical information on which the arguments in this paper are based has been extracted from a large number of publications and many of the accounts are not sufficiently detailed to enable the reader to assess the full significance of their content. Nevertheless, there are now a sufficient number of well-authenticated results available, for which full experimental details have been given, to provide overwhelming evidence that the remanent magnetic vectors of rocks do in fact change in a systematic way with geological age, and even after allowing for a possible subjectiveness of some workers in the rejection of data on certain occasions the plots in figure 7 evidently demonstrate the existence of a definite phenomenon, whatever its explanation may be.

3·4. *Statistical analysis of the values of* ϕ

A statistical analysis of the values of ϕ has been made and the results are set out in table 6 A. Column II of this table shows the total number n of pairs of values of

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T and ϕ for each of the five continental land masses, and column V gives the correlation coefficient between T and ϕ defined by the usual expression:

$$r = \frac{\Sigma(\phi_i - \overline{\phi})\,(T_i - \overline{T})}{\sqrt{[\Sigma(\phi_i - \overline{\phi})^2\,\Sigma(T_i - \overline{T})^2]}}, \tag{6}$$

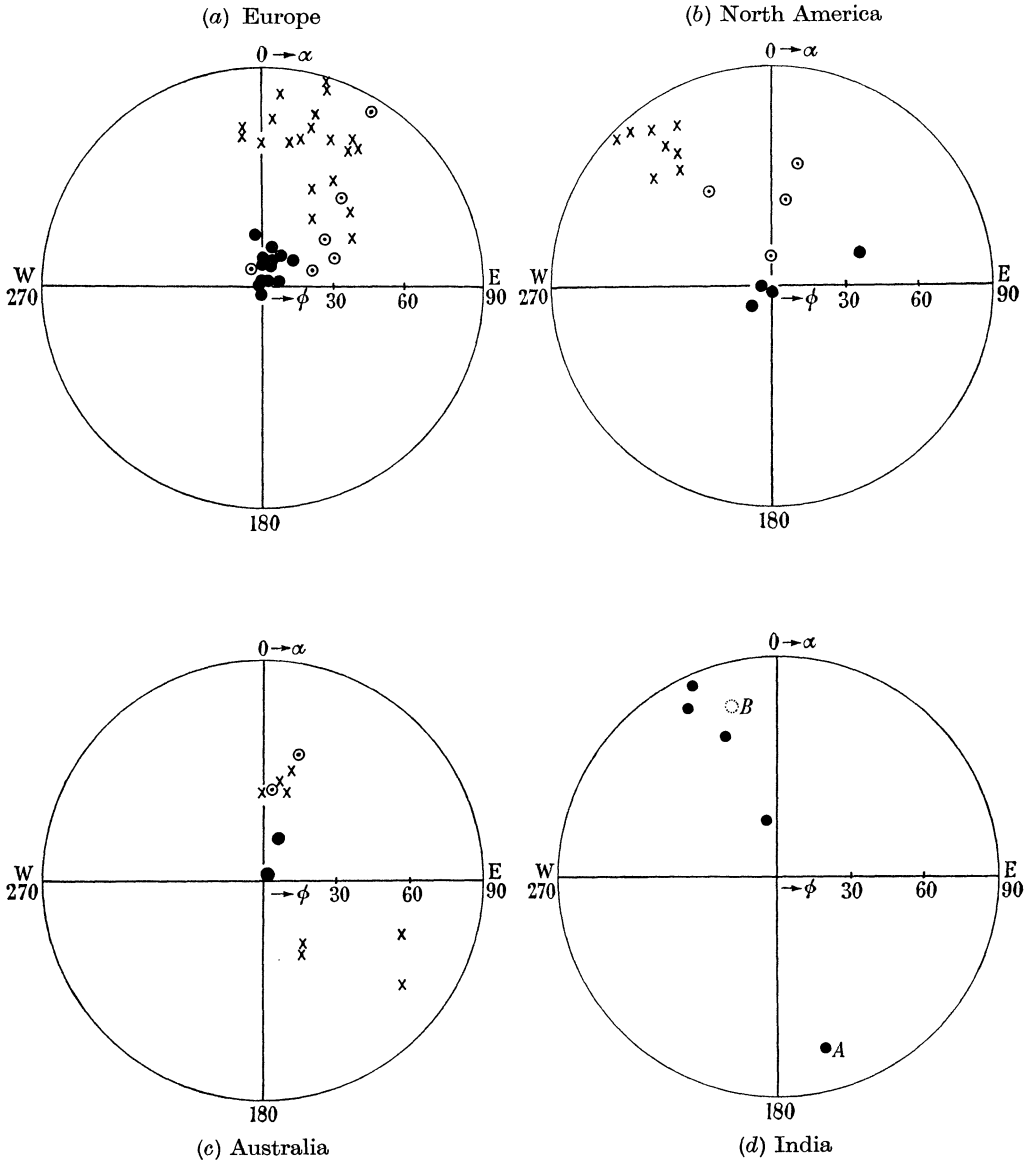


FIGURE 7. ●, Cenozoic; ○, Mesozoic; ×, Palaeozoic. (All north poles on lower hemisphere except B.)

where $\overline{T}$, $\overline{\phi}$ are the mean values for all sets of observations and T_i, ϕ_i, are the values of the ith pair. The reliability of r can be estimated from the precision factor p, defined as the probability that a completely random distribution would show equally good apparent correlation. The amplitude of p depends both on r and n,

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and values obtained from the statistical tables compiled by Fisher & Yates (1953) are given in column VI. In no case except that of Africa does p exceed 1 % so that we may assume the correlation coefficients to have high reliability for at least four of the continents.

It appears therefore that for Europe, North America, Australia and India we are justified in concluding that ϕ on the whole has been descreasing fairly steadily with time during the period we are considering. The data are as yet too scanty and the accuracy of the angle and age determination is too small for it to be possible to establish the finer details of the variation of the divergence with age. It may well be that over and above the rather steady increase shown in figures 2 to 5 further measurements may reveal marked oscillation or irregularities.

TABLE 6A

I	II	III	IV	V	VI	VII	VIII	IX	X
	n	$\overline{T}$ (My)	$\overline{\phi}$	r	p	$b=\mathrm{d}\phi/\mathrm{d}T$ (deg/My)	δb	ϕ_0	$\delta\phi_0$
Europe	43	146	37°	0·93	$<0{\cdot}01$	0·24	0·02	+3°	3°
N. America	16	196	48°	0·81	$<0{\cdot}01$	0·21	0·04	+7°	8°
Australia	12	233	36°	0·78	$<0{\cdot}01$	0·10	—	—	—
India	5	81	70°	0·99	0·01	0·89	—	—	—
Africa	8	186	47°	0·69	0·06	0·42	—	—	—

TABLE 6B. EUROPE

all rocks	43	146	37°	0·93	$<0{\cdot}01$	0·24	0·02	—	—
reversed	21	128	38°	0·903	$<0{\cdot}01$	0·27	0·03	—	—
normal	10	143	30°	0·608	0·06	0·11 (0·21)*	—	—	—
$N+R$	12	161	42°	0·876	$<0{\cdot}01$	0·26	—	—	—
igneous	29	112	30°	0·936	$<0{\cdot}01$	0·25	0·02	—	—
sedimentary	11	212	48°	0·537	0·1	0·14 (0·23)*	—	—	—

* These alternative values for b are obtained by joining the point $\overline{T}$, $\overline{\phi}$ to the origin.

However, in most cases the scatter of the points is so large that the best that can be done at present in the way of a preliminary numerical assessment of the data is to assume a linear relationship between ϕ and T. To establish such a relationship we have assumed a linear regression of ϕ on T of the form

$$\Phi = \overline{\phi} + b(T - \overline{T}). \tag{7}$$

This is represented in the diagrams of figure 4 by the straight lines which pass through the points $\overline{T}, \overline{\phi}$, and cut the ϕ axis at the point ϕ_0 given by

$$\phi_0 = \overline{\phi} - b\overline{T}. \tag{8}$$

The slope b is the coefficient of regression and is given by

$$b = \frac{\Sigma(\phi - \overline{\phi})\,(T - \overline{T})}{\Sigma(T - \overline{T})^2}. \tag{9}$$

The use of this method of determining the best straight line is, of course, somewhat arbitrary, but it is probably the most satisfactory way of treating the rather

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meagre results so far available, and we will therefore tentatively assume that the value of b defined in this way gives the most reliable estimate of $\mathrm{d}\phi/\mathrm{d}T$.

If n is sufficiently large the sampling error δb in the slope b is (Fisher 1954) given by

$$(\delta b)^2 = \frac{\{1/(n-2)\}\,\Sigma(\phi-\overline{\phi})^2}{\Sigma(T-\overline{T})^2}, \tag{10}$$

where ϕ is given by equation (4). If we assume the statistical sampling error $\delta\phi_0$ of ϕ_0 to be due solely to the sampling error δb of the slope we have

$$\delta\phi_0 = \overline{T}\delta b. \tag{11}$$

It is evident from table 6 that for the continents for which the data are sufficiently complete for statistical analysis, the values of ϕ_0 are comparable with $\delta\phi_0$ so that the fact that the regression lines do not pass through the origin is of no statistical significance.

We therefore conclude that the palaeomagnetic results obtained to date at least for the continents of Europe and North America are compatible with the supposition that the deviation ϕ has decreased approximately linearly with geological time T from values approaching or greater than 90° in the remote past to zero at the present time. The values of $\mathrm{d}\Phi/\mathrm{d}T$ for Europe and North America are of the order of 0·24 and 0·21 deg/My, respectively. The rate of change for India is considerably greater, the value of $\mathrm{d}\Phi/\mathrm{d}T$ being 0·89 deg/My years or so, but for this continent and Austrialia the number of results is insufficient to justify the use of equation (10) for estimating δb. For Africa the results so far are insufficiently conclusive to yield any reliable numerical value, and will not be further considered in this paper.

In table 6B are given for Europe only—sufficient data are lacking elsewhere—the values of $\mathrm{d}\Phi/\mathrm{d}T$ as determined from rocks magnetized both normally (N), rocks magnetized reversely (R) and rocks magnetized both normally and reversely at the same neighbouring sites ($N+R$).

Also included are the results of numerical analyses carried out separately on all igneous rocks and all sedimentary rocks from Europe. It is of particular interest to compare the values of $\mathrm{d}\Phi/\mathrm{d}T$ for these separate categories, since a number of the influences tending to change the direction of magnetization from that of the original magnetizing field must operate differently for rocks of these different classes (see also, §4·1). It is evident that for the reversed (R), normal with reversed ($N+R$), and igneous rocks the value of $\mathrm{d}\Phi/\mathrm{d}T$ does not differ markedly from that of the combined European results shown in table 6A. For normal (N) and sedimentary categories the slope is much less, but on the other hand, the correlation is considerably lower, and the errors δb are higher. If the origin ($\phi = 0$, $T = 0$) is taken as a point then the values of b, shown in brackets in table 6A, are quite close to the mean.

It must be remembered that the dates assigned to the rocks are their geological dates of formation, but not necessarily their dates of magnetization, which may in some cases be considerably later. Owing to chemical processes including partial metamorphism or to the effects of viscous magnetization, a rock may acquire a

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magnetization appropriate to some date later, but not earlier than its age of formation. Thus it is possible that some of the points in the figures should be shifted to lower values of T. It follows that this possible source of error will lead to the slopes of the curves ($\mathrm{d}I/\mathrm{d}T$, $\mathrm{d}\psi/\mathrm{d}T$, $\mathrm{d}\phi/\mathrm{d}T$) in figures 2 to 5 being too low. There is some reason to believe that sedimentary rocks may be more liable to this form of error than igneous rocks. If this were the case we would expect $\mathrm{d}\Phi/\mathrm{d}T$ for sedimentary rocks to be less than that for igneous rocks. For Europe we have seen that this is so. However, the difference is not statistically reliable. One can see that this type of dating error could explain the points in, say, the plots of ϕ against T which lie well below the mean regression line, but cannot explain the high points.

4. Possible explanations of the observed results

The deviation of ΔI, ψ and ϕ from zero in ancient times can be attributed to one or more of the following mechanisms:

(*a*) The direction of magnetization of the rock today does not represent the direction of the local earth's field at the time of magnetization.

(*b*) The main geomagnetic field has always been that corresponding to a dipole parallel to the rotational axis, but the earth's crust as a whole has shifted relative to this axis: this is the hypothesis of geographical polar wandering.

(*c*) The earth's crust has remained rigid and fixed relative to the present geographical poles, but the main earth's field has in the distant past corresponded to a dipole which was not directed along the rotational axis: this is the hypothesis of magnetic polar wandering.

(*d*) The earth's field had strong non-dipole components in remote geological times.

(*e*) Continental drift has occurred.

Many workers in palaeomagnetism support postulate (*e*). Discussion of this will be reserved until later, and for the present we will consider the possible implications of the results on the tentative assumption that the main land masses have remained in the same positions relative to one another at least since Cambrian (500 My) times. This includes the hypotheses (*a*), (*b*), (*c*), (*d*).

4·1. *Lack of parallelism between the direction of magnetization and the local field, hypothesis* (*a*)

The possibility that some large-scale geological or geophysical agent has produced the observed deviation of the remanent vectors of older rocks from that of an axial dipole field cannot be completely excluded. However, if any such process had in fact occurred, it must have operated over areas of continental extent and its effect must have caused a progressive variation over the past 500 My. Widespread mechanical stresses associated with orogenic movements in the past appear to provide the most likely mechanism of this kind suggested so far (Graham, private communication).

The approximate numerical agreement shown in table 6B between the values of $\mathrm{d}\Phi/\mathrm{d}T$ for different classes of rocks (igneous, sedimentary, etc.) taken separately

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makes it unlikely that the non-zero values of ϕ are critically dependent on the detailed structure of the magnetic materials in the rocks. For igneous and sedimentary rocks are in general extremely unlike in physical and chemical structure and, moreover, are generally magnetized in quite different ways: igneous rocks by thermomagnetic remanence and sedimentary rocks mainly by chemical magnetization or occasionally during deposition. Igneous rocks are on the average a thousand times as intensely magnetized as sedimentary rocks.

If further measurements confirm with greater precision the equality of the values of ϕ for rocks which have been thermoremanently magnetized and for those nagnetized in other ways, of the same age from the same neighbourhood then one can be certain that the magnetization of neither has been markedly and consistently altered from that of the ambient field at the time of formation, and one will be able to exclude completely mechanism (a) as the major cause of the non-zero value of ΔI, ψ and ϕ. For the rest of the discussion this mechanism will be assumed excluded.

4·2. *The geographical polar wandering hypothesis, hypothesis* (b)

According to the polar wandering hypothesis the whole of the earth's crust together with the continental land masses, has shifted as a rigid whole relative to the rotational axis which has remained fixed relative to the stars. It is convenient to think of this as a movement of the poles over the surface of a fixed earth. It is now almost a century since the suggestion of polar wandering was first put forward, but the theoretical aspects of the problem have recently been re-examined by Gold (1955) and by Munk (1956), who concluded that it may well occur as a result of such processes as the formation and melting of ice caps, or mountain building.

If the earth's field has always been that of an axial dipole, and if the increase in the value of ΔI, ψ and ϕ for older rocks is due to polar wandering in the past, then one would expect that the geographical positions of the north and south poles of the earth, as calculated from the values of I and ψ, for rocks for different lands, would be the same at any one geological epoch.

The practice of expressing rock magnetic results in terms of ancient pole positions was first introduced by Creer *et al.* (1954) and has since been widely adopted. Corresponding to any pair of observed values of I and ψ, for rocks collected from a locality whose present latitude and longitude are λ_0, and I_0, respectively, the latitude λ_p, and longitude L_p of the magnetic pole are defined by

$$\sin\lambda_p = \sin\lambda\sin\lambda_0 + \cos\lambda\cos\lambda_0\cos\psi, \tag{12}$$

$$\sin(L_p - L_0) = \frac{\sin\psi\cos\lambda}{\cos\lambda_p}, \tag{13}$$

where λ is the ancient magnetic latitude of the formation at the time when the rocks acquired their magnetization, and is given by

$$\tan\lambda = \tfrac{1}{2}\tan I. \tag{14}$$

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Irving (1959) shows, as many other workers have done, that there are appreciable differences between the polar wandering curves calculated from data for the different continents; for geological epochs dating back to more than 50 My the discrepancies are large. It appears therefore that while polar wandering may have occurred in the remote past, we cannot account for the palaeomagnetic results by this hypothesis alone.

4·3. *The inclined dipole field hypothesis or magnetic polar wandering, hypothesis* (*c*)

Suppose that the main geomagnetic field has always approximated to that of a dipole whose axis has been inclined to the earth's rotational axis at an angle varying slowly with time. Again one would expect the geographical positions of the magnetic poles, as calculated from I and ψ, for rocks of all land masses, to agree at the same geological epoch. It is evidently not possible to distinguish by palaeomagnetic measurements between this hypothesis and that of geographical polar wandering discussed in §4·2 above.

4·4. *The non-dipole field hypothesis, hypothesis* (*d*)

A possible explanation of the observed discrepancy between the contemporaneous pole positions would be to suppose that the earth's field departed widely and consistently from that of a dipole for long periods of time. The non-dipole part of such a field would have to have been of markedly larger amplitude and must have changed with time vastly more slowly than the secular variation field of the earth in the present epoch. The latter field has a maximum amplitude at the surface of not more than 30 % of the equatorial surface field H_0, and a period of change of the order of a few thousand years. To explain the results shown in figures 2 to 5 by a non-dipole field, one would have to assume a somewhat larger maximum amplitude, at least as large as H_0, and a period of change of the order of a few hundred million years, that is 10^5 times longer.

The results shown in figures 2 to 5 could be explained on this hypothesis. All that the measurements reliably give us to date is a rough estimate of the mean directions of the field over four continents as a function of geological age. It would clearly be easy to choose arbitrarily a set of time-varying multipoles which would reproduce the observations.

In principle it should be possible to test this by comparing the directions of remanent magnetism of rocks of the same age taken from widely separated sites on a single assumed rigid land mass: for, if the ancient field had markedly multipolar components, one would expect the rate of change with distance on the earth's surface to be more rapid than for a simple dipole field, and directional differences should be detectable over the area of a large continent.

To be sure, this conclusion does rest on an assumption which cannot yet be directly proved: this is that the nature of the assumed non-dipole field is in no way related to the distribution of continents and oceans on the earth's surface. Any theory which places the origin of the field primarily in the earth's core will satisfy this condition: all tenable current theories do this.

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To make such a test, rather precise measurements of the ancient field are needed at sites which are very far apart on a single land mass. So far the measurements in North America are the only ones which begin to approach but do not, in fact, yet meet this requirement. Consider, for instance, sites numbered 6, 8 and 15 (figure 1) near the east coast of the U.S.A. These are about 2000 miles from the more numerous sites near the west coast. In figures 3(*a*) and (*b*) the three east-coast points are marked with a circle and it will be seen that they lie close to the mean curve. Thus no difference between the field direction defined by ΔI and ψ between and east and west coast can be demonstrated. It may be noticed from figures 2(*a*) and 3(*a*) that at $T = 200$ to 300 My there is little difference between the values of ΔI for North America and Europe, which are almost 4000 miles apart, so that the absence of an established difference between the east and west of North America is not surprising. On the other hand, the corresponding mean values of ψ for America and Europe are about $-25°$ and $+25°$, respectively, giving a difference of 50°. So the absence of a difference of ψ between east and west of North America would be significant if firmly established statistically. Unfortunately this cannot be said to be so. In this argument it has been assumed that the geological foldings resulting in the Appalachians and Rocky Mountains have not appreciably distorted the geometry of North America.

Another and, in principle, more precise method is to compare the rate of change with distance apart of the pole positions calculated for sites within a single continent with the corresponding rate of change for different continents. Irving has calculated (figure 20 of his 1959 paper), the distribution of measured angles between pairs of pole positions calculated (*a*) for the sites within one continent, and (*b*) for sites in different continents, for the period from about 200 to 50 My. The former mean angle is about 15° and the latter about 60°. Thus the actual difference between the pole positions is some four times greater between, compared with within, the continents. However, the average distance apart of the continents considered is much larger, by perhaps ten times, than the average distance apart of the sites within each continent. Thus, within the very limited accuracy of the data, the rate of change with distance of the difference between the pole positions within and between the continents is about the same. Thus the present data, as analyzed by Irving, are consistent with the hypothesis of the non-dipole field.

In general then, one can conclude that it is not at present possible to exclude, on phenomenological grounds alone, the possibility that the rock magnetic data now available might be explained by the two assumptions of a time-dependent non-dipole field together with the fixity of the continents.

4·5. *The hypothesis of continental drift, hypothesis* (*e*)

If the arguments given above are held to be adequate to exclude mechanisms (*a*), (*b*) and (*c*) as the main explanation of the observed results, one is left with (*d*) the non-dipole field hypothesis and (*e*) continental drift, as the two remaining possibilities. In the next section, the latter hypothesis will be examined in detail.

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5. Rock magnetism and continental drift

5·1. *General considerations*

In accordance with the foregoing discussion we will make the tentative assumption that the original mean direction of magnetization of all rocks corresponded to that of the local axial dipole field, and that the non-zero mean values of ΔI and ψ for large T are due to a movement of the continents relative to one another since the remanent magnetism was acquired. The problem is to ascertain precisely what can be deduced quantitatively about this postulated drift from the curves of $I(T)$ and $\psi(T)$ in figures 2 to 5, (a) and (b).

It has become customary in discussing palaeomagnetic results to determine from I and ψ the latitude and longitude of the ancient magnetic pole for each continent and each geological epoch. If the pole positions calculated for all continents at each age in the past had coincided within the limits of experimental error, with one another but not with the present geographical pole, then one would have concluded that the data could be explained by magnetic polar wandering alone, and that no other mechanism need be invoked. We have seen that there is in fact no such coincidence, and that this has led many workers to suggest that continental drift has occurred. The now widespread use of pole positions in discussing quantitatively the problem of continental drift, however, leads to needless complication. This is evident when we consider in an elementary way the basic principles of the method, which have been given in detail in previous sections. In essence they are as follows.

(a) From the measured values of I and ψ made on rocks from a given site S, the ancient latitude λ of the site is calculated. The quantities ψ and λ then define the ancient pole position *relative to* S—that is with S as the origin of co-ordinates. One can visualize this simply by imagining a globe of the earth as it appears at the present time, with a wire bent into the arc of a great circle, and rigidly attached at S in such a way that it makes an angle ψ with the present meridian passing through that point. If the length of the wire corresponds to the ancient co-latitude, $90° - \lambda$, its farther end, P, will correspond to the ancient magnetic pole position.

(b) The position of P is now found in the present-day co-ordinate system; that is, with the present equator and the Greenwich meridian as fiducial lines.

(c) A second ancient magnetic pole position P' is calculated in present-day co-ordinates in the same way from data obtained for rocks of the same age at a point S' in another continent.*

(d) The earth's field is assumed to have been that of an axial dipole so that P and P' are also taken to correspond to the ancient *geographical* pole. This means that at the epoch when the rocks were laid down these two points were coincident. To find what we can about the ancient relative positions of the two continents we must envisage them on our model as being moved from their present situations

* In practice it is impossible to date two rock formations in separate continents with sufficient accuracy to carry out this simple procedure, and it is usual to draw out a curve for each continent showing how the pole positions have changed with geological time. This, however, does not in principle affect the present argument.

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and made to slide over the surface of the globe, still carrying the wires SP and $S'P'$ rigidly attached, until P' falls on P. Now there are an infinite number of ways in which this can be brought about, for once P' coincides with P, both S and S' can evidently be moved around in small circles with this common point as pole.

In discussing the geological implications of rock magnetic measurements it is in many ways less confusing to consider the actual geographical latitudes and orientations, in so far as they can be determined, of the various land masses during each particular epoch of the past rather than to think in terms of ancient pole positions; for whatever changes may have occurred during remote ages the rotational axis has always been the one constant unchanged feature by which the position of an object on the earth's surface can be defined. This approach will be adopted in the quantitative discussion given in the following section.

5·2. *Numerical results*

The only rock magnetic data obtainable to establish the ancient geographical position of each individual site within a single continent are the declination D and the inclination I. From these we can find directly the rotation ψ and, from equation (14), the ancient latitude λ. As indicated in §4·4 the available data are as yet insufficient to establish significant differences between D and I within the same continent. At present the best that can be done is to use the measurements at all the different sites to give a mean value of λ and ψ for the continent as a whole.

Table 7. Reference places in the continents

I_0 is magnetic inclination given by $\tan I_0 = 2 \tan \lambda$.

continent	place	latitude	longitude	I_0
W. Europe	Paris	49 N°	2 E°	67° down
N. America	Denver	40 N°	105 W°	59° down
India	Nagpur	22 N°	79 E°	39° down
Australia	Alice Springs	23 S°	134 E°	40° up

To obtain the mean latitude of a continent it is convenient to select some central town to act as a geographical reference point. The towns selected, with their present geographical latitude λ_R and longitude L_R, are given in table 7. Then from the measurements of I and ψ at a site, situated at present geographical latitude λ_0 and longitude L_0, the corresponding pole positions at λ_p and L_p is calculated from equation (12) and (13). Then, from this pole position, the ancient latitude λ_m of the reference town at a given date is calculated from the relation

$$\sin \lambda_m = \sin \lambda_p \sin \lambda_R + \cos \lambda_p \cos \lambda_R \cos (L_R - L_p). \qquad (15)$$

Thus each site in the continent yields one determination of the ancient latitude of the reference town. The values obtained in this way for the towns of Paris (Europe), Denver (North America), Alice Springs (Australia) and Nagpur (India) are plotted in figures 2 to 5, (*d*),* and the curve through them gives the best available estimate of λ_m as a function of geological age. With the present low

* This has been done previously by Irving for Australia only; he used a different reference point situated in the south-east of the continent.

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accuracy of the data it is adequate to take the simple mean of the orientation ψ at each site to give the best value of ψ as a function of T for the continent as a whole. The curves of figures 2 to 5, (*b*), give this required orientation. If greater precision is needed a similar method to that used above for the latitude can be employed.

In table 8 are given the values of latitude λ_m of the reference town and the orientation ψ of each continent, for which there is adequate data. Those values which are especially uncertain are shown in brackets. These figures represent in fact all that rock magnetic data can tell about the drift of the continents on the assumption of an ancient axial dipole field.

TABLE 8. ANCIENT LATITUDE λ_m AND ORIENTATION ψ OF FOUR CONTINENTS AT VARIOUS GEOLOGICAL AGES

age (My) ...		0	50	100	150	200	300	400	500
Europe	λ_m	49° N	40° N	33° N	23° N	15° N	1° S	20° S	36° S
	ψ	0°	+10°	+19°	+23°	+27°	+27°	+18°	0
N. America	λ_m	40° N	33° N	24° N	20° N	15° N	2° N	4° S	8° S
	ψ	0°	(+20°)	(+16°)	+3°	−16°	−35°	−46°	−54°
India	λ_m	22° N	4° S	28° S	56° S	—	—	—	—
	ψ	0°	−15°	−28°	−32°	—	—	—	—
Australia	λ_m	23° S	40° S	57° S	62° S	65° S	19° S	7° N	11° N
	ψ	0°	+20°	(+33°)	(+40°)	(+51°)	(+59°)	+57°	+46°

Figures in brackets are subject to more than the average uncertainty; see figures 3(*b*) and 4(*b*).

In figure 8 the points show the geographical positions, derived from rock magnetic data, of Paris, Denver, Alice Springs, and Nagpur, for past ages extending back to 500 My. These have been obtained by putting the reference town of each of the four continents at the appropriate latitude λ_m for each epoch. The arrows represent a rotation of $-\psi$, which brings back the orientation to its ancient value relative to the north geographical pole of the past.

5·3. *Some conclusions*

Some of the most striking results are as follows: all four of the land masses studied have, during the last 200 My, apart from possible short-lived retrogression, been moving steadily northwards with velocities between 0·2 and 0·8° of latitude per million years.

In the last 300 My the mean latitudes of western Europe and North America have changed in almost the same way, from near the equator to their present positions of about 50° N. This common movement in latitude is in sharp contrast with the different movement in azimuth; Europe has rotated about 50° clockwise relative to North America. Australia appears to have moved in latitude in a somewhat complicated way. It appears to have been near the equator 400 My ago, then to have drifted south to within about 20° of the pole at 200 My, and afterwards northwards again to its present latitude of 25° S. Although the data are still scanty, India has apparently moved farther and faster than any other continent, from 50° S. at T = 150 My to the equator at T = 80 My and then to its present latitude of 20° N.

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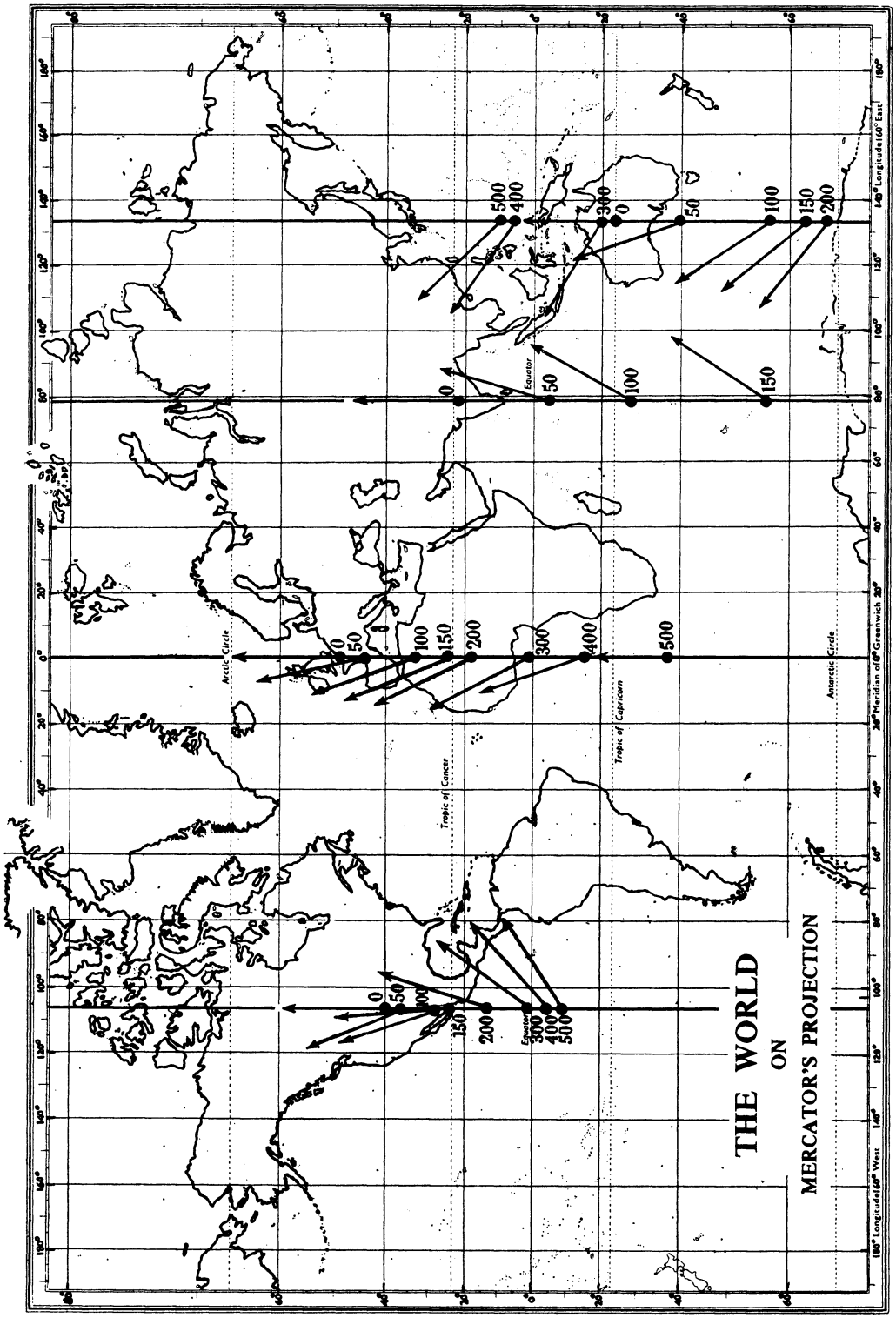


FIGURE 8. The number against each point indicates its age in millions of years.

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Turning now to the question of longitude, in figure 8 the positions of reference towns have been adjusted in such a way that each lies on its present meridian. Thus, to take an example, Denver is shown to be on the line of longitude 105° W. of Greenwich for all ages. It is apparent, however, that the relative longitude of Denver and, say, Paris, are indeterminable for past geological epochs; for all that can be found from the measured values of D and I are the values of ψ and λ_m, respectively, and these are essentially independent of longitude on the dipole field hypothesis. It is obviously impossible by purely physical measurements alone, to determine the past relative *longitudes* of two land masses in the past. To take a specific example, we have no way of ascertaining by rock magnetic evidence whether America has moved further from or nearer to Europe during the past few hundred million years.

The palaeomagnetist is in exactly the same situation as a seaman only able to observe the altitude of the pole star. He can deduce the orientation of his ship and its latitude but can say nothing about its longitude. In the same way a palaeomagnetist can determine the ancient latitude λ of a site and its orientation relative to the ancient meridian, but the ancient longitude is indeterminate.

6. Discussion

6·1. *The use of theoretical arguments involving knowledge of the structure of the earth*

If the arguments presented in the foregoing sections are accepted one can conclude that the present rock magnetic data may be explained either by (*a*) the directions of magnetization of older rocks differing from that of the original magnetizing field, (*d*) the existence of a non-dipole field in the past, (*e*) continental drift.

(*a*) It is easy to envisage a number of possible processes which might change the magnetic directions in rocks over geological time. The effect of widespread tectonic forces has already been mentioned as one possibility. However, it is difficult to envisage any such agent which would produce a uniform trend in magnetic direction over continental areas, and we will tentatively suppose the continuous increase of ϕ with T to be explained either by a non-dipole field without drift, or by drift and a dipole field, or some combination of these.

There are, of course, many theoretical suppositions involving assumptions about the structure of the earth, which can be involved to give to or withhold support from both the multipolar field and drift hypotheses (*d*) and (*e*). This is not, however, the place to discuss such theories in detail. For, on the one hand, the theoretical arguments for and against continental drift are so uncertain that they cannot at present in our view be usefully employed either to support or refute the hypothesis.

As regards the suggestion of a non-dipole field, there are indeed many good theoretical reasons why large multipolar fields are unlikely to have persisted for periods of the order of 100 My. This is because almost all theories of the earth's field explain it as due to convection in the liquid core, where the characteristic time scale for change would appear to be rather short, and only of the order of a few

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thousand years or so. However, our knowledge of the core is so slight that no theory at present can be considered as established. As an indication of the uncertainty about the nature of the core, it appears neither agreed whether the assumed convection is due to radioactivity or to chemical change, nor whether a small solid core exists within the larger liquid core. So long as such basic uncertainties exist in our knowledge of the interior of the earth, it is as well to put little weight on deduction from any particular theory. We conclude that theoretical arguments at present are not sufficient to distinguish between the two hypotheses.

Since the field of a dipole falls off as $1/r^3$ and that of a harmonic of degree n as $1/r^{n+1}$, one would need very large higher harmonics at the core to give a non-dipole field as large as the dipole field at the surface. This might be considered as a difficulty for the non-dipole field theory were it not for the fact that the theory of the earth's dipole field which is today perhaps the least implausible, does imply large toroidal fields (~ 400 Oe) at the core: if toroidal fields of this magnitude exist, then one cannot rule out multipole fields of the same strength.

There is, however, one general phenomenological argument which can, perhaps, be usefully invoked. On any hypothesis which invokes a field differing from that of an axial dipole, the present epoch must be unique in geological history; whereas on the drift hypothesis, the present epoch may well be typical of all time.

The reasoning is as follows. Since one sees from the curves of figures 2 to 5, (a) and (b), that the mean values of ΔI and ψ are nearly zero for all continents during the last, say, 20 My, it follows that the average values taken over a few thousand years of any non-dipole field must have been quite small, perhaps not more than a few parts per cent of the present dipole field. On the other hand, at epochs earlier than 200 My, the generally large values of ΔI and ψ show that the ancient non-dipole field must have been comparable with the present dipole field; so that the explanation of the rock magnetic results by the non-dipole hypothesis leads one to the conclusion that the small value of the non-dipole field is peculiar to the present age.

On the other hand, the situation with regard to the drift theory is quite different, for there is no indication in figures 2 to 5 that the slopes of the curves of ΔI and ψ, against T differ markedly at the origin from say 300 My ago. Thus if we assume an axial dipole field, and interpret the palaeomagnetic results obtained so far as being due to continental drift, it seems plausible to assume on the basis of the present data that the rate of movement of the main land masses today is very much the same as it has been throughout the geological history of the earth.

6·2. *Comparison of deductions from rock magnetic data with other observed facts*

To summarize briefly the conclusions arrived at in this paper, it seems that the general increase of the divergence ϕ with T can, on the basis of our present knowledge, be explained in any one of the three ways, (a), (d) and (e) outlined in §6·1.

If hypotheses (a) or (d) are assumed to be the explanation of the magnetic measurements, then there are not now, nor soon likely to be, any sets of observational data other than those of rock magnetism, which can be used directly to test these hypotheses. For few, if any, physical effects, other than the magnetic phenomena

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themselves can be expected to be observable at the surface of the earth. Methods by which, in principle, given more and better magnetic data, hypothesis (*d*) could be confirmed or refuted, have already been outlined in § 4·4.

A quite different situation exists in relation to the hypothesis that the magnetic data are to be explained by invoking drift. For historically the suggestion of continental drift long preceded rock magnetic studies. It was based entirely on a multitude of terrestrial surface phenomena, geometrical, geological, palaeontological, and palaeo-climatological. Accepting for the present what seems to be the general view that such data alone neither conclusively support nor refute this hypothesis, the obvious procedure is to test numerically the quantitative deductions made from palaeomagnetic measurements which are summarized in table 8, against the other set of facts. Without doubt, the most suitable of these is the climatological evidence. For the terrestrial climate is now, and to some degree must always have been, markedly zoned according to geographical latitude. Now, it is just the geographical latitude which is the main quantity yielded by the rock magnetic data, on the assumption, of course, of an axial dipole field. Thus it is clearly possible to test in a numerical manner whether the rock magnetic latitudes are or are not consistent with the ancient climates. This comparison will be made in another paper, and will be based on the ancient latitude and orientation of a given locality, as calculated from the ancient latitude and orientation of the reference town, as tabulated in table 8.

The authors are indebted to many colleagues for useful discussion and criticism: most particularly to Professor G. A. Barnard, Professor J. D. Bernal, F.R.S., Professor M. Blackman, Sir Edward Bullard, F.R.S., Dr W. Bullerwell, Dr H. Elliot, Dr John W. Graham, and Professor P. Morrison. They are also very grateful to Mr G. W. Massé for considerable assistance in drawing up the diagrams and tables.

Appendix I. Rock magnetic reversals and the choice of ϕ.

One difficulty arises owing to the existence of rock magnetic reversals. Of numerous samples of geologically recent rocks collected from different localities in the northern hemisphere, it has been found that while their mean axis of magnetization corresponds fairly closely to that of an axial dipole field, approximately half are reversely magnetized. Detailed investigations have shown that if at any particular locality a succession of rocks of ages ranging over many thousands of years are taken from a single formation they may be either

(1) entirely normally magnetized (that is, towards the north);
(2) entirely reversely magnetized (that is, towards the south); or
(3) some specimens may be normal and others reversed in magnetic direction.

Two explanations have been put forward to account for the occurrence of reversely magnetized rocks. (*a*) It has been suggested that the main geomagnetic field itself underwent a sequence of sudden reversals at intervals of the order of a million years. (*b*) Alternatively, it may be that many rocks due to rather complex physico-chemical processes became magnetized in a direction opposite to that of the earth's field either at the time when they were formed or during some later epoch.

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Turning now to older rocks, whose remanent magnetic vectors diverge widely from that of a present-day axial dipole field, we again find that for a certain limited number of geological formations, samples taken from different places show two distinct magnetic polarizations, differing in direction by approximately 180°. It appears, therefore, that one or both of the reversal processes (*a*) and (*b*) may have occurred at least during certain periods, throughout geological time. This evidently means that if a particular rock mass became magnetized along the ambient field of the earth during some epoch of the remote past, palaeomagnetic measurements made on it can give us only the orientation of the axis of the field vector at that time, and can tell us nothing about its sense. There may arise in certain cases an inherent ambiguity of 180° in the values of ψ and I and therefore also an ambiguity of ϕ and α, and two alternative positions for each of the plots in figures 2 to 5, (*c*), and 7. The existence of this ambiguity represents a source of uncertainty in the interpretation of the results for some of the older rocks.

In constructing figure 7 the criterion adopted was that when any doubt arises the position of the point should be chosen so as to follow most closely the obvious general trend. There is in fact only one point that is seriously affected. This is the plot for India at 150 My mentioned previously in § 3·2. The two alternative positions are shown in figure 7 (*d*) by the dot A and the open circle B. Point A represents a vector directed downwards into the plane of the paper with $\phi < 180°$ as do the other four points in the diagram, while for the antipodal position B the vector is directed upwards with $\phi > 180°$. It is evidently logical to assume that B denotes the true direction, since this places the four plots on the shortest smooth curve.

This criterion for the resolution of the sense ambiguity is evidently identical to that given in § 3·2 that the position should be chosen so as to avoid large abrupt changes in $\mathrm{d}\phi/\mathrm{d}T$. It also implies that $\mathrm{d}(\Delta I)/\mathrm{d}T$ and $\mathrm{d}\psi/\mathrm{d}T$ should change least abruptly.

In tables 1 to 5 and throughout the text the convention used for distinguishing between normal (N) and reverse (R) magnetization is that the rocks should be listed as reversed when and only when the actual direction of remanent magnetization, defined by the original measured quantities D and I, makes an angle greater than 90° with that of the present dipole field at the site where the rocks were found.

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Appendix II. Tables 1 to 5

Key to tables 1 *to* 5

I	geological formation
II	igneous/sedimentary
III	T = geological age (My)
IV	λ_0 = present latitude of site L_0 = present longitude of site
V	I_0 = present dipole inclination
VI	N = number of specimens
VII	D = declination of remanent vector I = inclination of remanent vector
VIII	δ = semi-angle 95 % cone of confidence
IX	ΔI = change in inclination
X	ψ = rotation of site
XI	λ_p = latitude of ancient pole L_p = longitude of ancient pole
XII	λ = ancient latitude of site
XIII	λ_m = past latitude of reference point
XIV	ϕ = angle of divergence
XV	α = deviation of remanent vector
XVI	whether normal (N) or reversed (R)
XVII	reference

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Table 1. Europe

	I geological formation	II ig sed	III T	IV λ_0	IV L_0	V I_0	VI N	VII D	VII I	VIII δ	IX ΔI	X ψ	XI λ_p	XI L_p	XII λ	XIII λ_m	XIV ϕ	XV α	XVI N or R	XVII reference
1	Mt. Etna lavas	I	<15	38° N	15° E	57° d	11	4°	56° d	2°	−1°	+4°	86° N	126° E	37° N	47° N	3°	48°	N	Chevallier (1925)
2	Central France lavas	I	<15	45° N	3° E	63° d	—	206°	60° up	—	−3°	+26°	78° N	93° E	41° N	47° N	13°	33°	R	Roche (1951)
3	Post-glacial SW flows* (Iceland Quaternary)	I	<15	64° N	21° W	76° d	—	1°	74° d	8°	−2°	+1°	86° N	151° E	60° N	46° N	2°	8°	N	Hospers (1955)
4	Flows W and SW* (Iceland Quaternary)	I	<15	64° N	22° W	77° d	51	181°	75° up	7°	−2°	+1°	88° N	147° E	62° N	47° N	2°	7°	R	Hospers (1955)
5	Normal flows* (Iceland Quaternary)	I	<15	65° N	22° W	77° d	—	26°	75° d	—	−2°	+26°	88° N	149° E	62° N	47° N	7°	60°	N	Sigurgeirsson (1957)
	Reversed flows* (Iceland Quaternary)	I	<15	65° N	22° W	77° d	—	180°	74° up	—	−3°	0°	77° N	74° E	60° N	51° N	3°	0°	R	
6	Mt. Dore and Limagne lavas	I	17	46° N	3° E	64° d	—	187°	50° up	—	−14°	+7°	73° N	167° W	31° N	33° N	15°	18°	R	Roche (1951)
7	Iceland N and NW Miocene flows	I	25	62° N	20° W	75° d	102	1°	78° d	6°	+3°	+1°	89° N	5° E	67° N	50° N	3°	176°	$N+R$	Hospers (1954)
8	Vogelsbergs lavas	I	25	50° N	9° E	68° d	42	8°	57½° d	6°	−10°	+9°	75° N	161° E	38° N	35° N	11°	26°	$N+R$	Angenheister (1956)
9	Limagne intrusive	I	40	46° N	3° E	64° d	—	180°	73° up	—	+9°	0°	77° N	3° E	59° N	62° N	9°	0°	R	Roche (1950*b*)
10	Gergovie intrusives	I	40	46° N	3° E	64° d	—	200°	53° up	—	−11°	+20°	73° N	119° E	34° N	39° N	15°	53°	R	Roche (1950*a*)
11	N England, NW dykes	I	40	55° N	3° W	71° d	—	174°	73° up	—	+2°	−6°	86° N	76° W	59° N	50° N	2°	299°	R	Bruckshaw & Robertson (1949)
12	Mull lavas	I	60	56° N	6° W	72° d	—	175°	77° up	24°	+5°	−5°	82° N	71° W	65° N	51° N	4°	334°	R	Bruckshaw & Vincenz (1954)
13	Mull intrusives	I	60	56° N	6° W	72° d	—	356°	52° d	—	−20°	−4°	72° N	139° E	33° N	35° N	20°	353°	$N+R$	Vincenz (1954)
14	Arran dykes	I	60	55° N	5° W	72° d	—	1°	64° d	—	−8°	+½°	79° N	154° E	46° N	39° N	8°	2°	$N+R$	Leng (1955)
15	Antrim lavas	I	60	55° N	6° W	71° d	—	188°	63° up	—	−8°	+8°	78° N	145° E	44° N	39° N	9°	24°	R	Wilson (1959)
16	Antrim lavas	I	60	55° N	6° W	71° d	24	194°	60° up	5½°	−11°	+14°	74° N	133° E	41° N	37° N	13°	33°	R	Hospers & Charlesworth (1954)
17	Lst and Sst of Skye and Brora	S	140±20	57° N	5° W	72° d	14	54°	66° d	6½°	−6°	+54°	56° N	76° E	48° N	47° N	20°	74°	$N+R$	Nairn (1957*a*)
18	L. and M. Jurassic Alps	S	140±20	48° N	13° E	65° d	21	37°	48° d	5½°	−17°	+37°	59° N	128° E	29° N	27° N	30°	54°	N	Hargroves & Fisher (1959)
							30	48°	51° d	6½°	−14°	+48°	53° N	112° E	32° N	28° N	30°	69°	N	
19	Keuper marl	S	170±20	53° N	2° W	69° d	540	33°	27° d	12°	−42°	+33°	43° N	131° E	15° N	12° N	46°	42°	$N+R$	Clegg *et al.* (1954)
20	Vosges sst	S	170±20	54° N	7° E	70° d	65	214°	11° d	11°	−81°	+34°	19° N	137° E	6° S	11° S	85°	33°	R	Stubbs (1958)
21	(Spain) sst	S	170±20	43° N	5° W	62° d	92	353°	57° d	2½°	−5°	−7°	82° N	131° W	38° N	43° N	7°	327°	N	Stubbs (1958)
22	Exeter volcanics	I	200±20	51° N	4° W	68° d	34	9°	9° d	20°	−59°	+9°	43° N	164° E	5° N	3° N	58°	11°	N	Creer (1957)
23	Mauchline sst+lavas	$I+S$	200±20	56° N	4° W	72° d	60	184°	5° up	10°	−67°	+4°	37° N	169° E	3° N	3° S	66°	4°	R	du Bois (1957)
24	Esterel flow R4	I	200±20	43° N	7° E	62° d	5	210°	1° up	—	−61°	+30°	46° N	141° E	½° N	11° N	65°	33°	R	Roche (1957)
25	Esterel dolerite	I	200±20	43° N	7° E	62° d	3	175°	2° d	—	−64°	−5°	43° N	165° E	1° S	13° N	64°	354°	R	Roche (1957)
26	Nideck porphyry	I	200±20	48° N	7° E	66° d	49	193°	7° up	4½°	−59°	+13°	43° N	168° E	4° N	3° N	60°	15°	R	Nairn (1957*b*)
27	Montcenis sst	S	200±20	47° N	4° E	65° d	14	197°	6° d	4°	−71°	+17°	38° N	162° E	3° S	1° S	72°	18°	R	Nairn (1957*b*)
28	Oslo Graben lava	I	200±20	60° N	11° E	74° d	12	202°	33° up	10°	−41°	+22°	46° N	165° E	18° N	6° N	42°	28°	R	Rutten *et al.* (1957)
29	St Wendel sst	S	200±20	49° N	7° E	66° d	27	181°	9° up	3½°	−57°	+1°	45° N	175° W	5° N	4° N	57°	1°	R	Nairn (1957*b*)
30	Esterel flow R3	I	200±20	43° N	7° E	62° d	14	217°	23° up	4½°	−39°	+37°	45° N	131° E	12° N	14° N	46°	50°	R	Rutten *et al.* (1957)
31	Pennant sst	S	250±30	51° N	2° W	68° d	14	44°	35° d	—	−33°	+44°	43° N	114° E	19° N	20° N	40°	62°	N	Clegg *et al.* (1954)
32	Clee Hill sill+contacts	$I+S$	250±30	52° N	3° W	69° d	63	200°	15° d	—	−84°	+20°	27° N	155° E	8° S	10° S	86°	19°	R	Everitt (1959)
33	Lundy granite	I	250±30	51° N	5° W	68° d	10	175°	9° up	27°	−59°	−5°	43° N	180° E	5° N	2° N	60°	354°	R	Blundell (1957)
34	Tideswelldale contacts	I	250±30	53° N	2° W	70° d	5	218°	36° d	—	−34°	+38°	5° N	143° E	20° S	26° S	70°	32°	N	Everitt (1959)
35	Shatterford intrusives	I	250±30	52° N	2° W	69° d	34	33°	3° d	—	−66°	+33°	32° N	137° E	2° N	0° N	68°	36°	$N+R$	Everitt (1959)
36	Derbyshire sst+lavas	$I+S$	250±30	53° N	2° W	69° d	—	26°	40° d	—	−29°	+26°	53° N	146° E	23° N	16° N	32°	39°	N	Belshé (1957)
37	Kinghorn lavas	I	250±30	56° N	3° W	72° d	102	24°	15°–42° up	—	−87°–114°	+24°	8° N–24° N	156° E	8° S–24° S	28° S	64°–90°	26°	$N+R$	Everitt (1959)
38	Anglo-Welsh cuvette	S	300±20	52° N	3° W	69° d	1081	201°	1° up	10°	−68°	+21°	30° N	159° E	2° N	8° S	66°	18°	R	Creer (1957)
39	Midland-Valley lavas	I	300±20	57° N	2° W	72° d	147	46°	54° up	—	−126°	+46°	10° S	141° E	35° S	38° N	49°	34°	$N+R$	Stubbs (1958)
40	Caerbwby sandstone	S	450±50	52° N	5° W	62° d	75	187°	39° d	8°	−23°	+7°	15° N	173° E	22° N	26° S	77°	6°	N	Creer (1957)

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TABLE 2. NORTH AMERICA

	I geological formation	II ig sed	III T	IV λ_0	IV L_0	V I_0	VI N	VII D	VII I	VIII δ	IX ΔI	X ψ	XI λ_p	XI L_p	XII λ	XIII λ_m	XIV ϕ	XV α	XVI N or R	XVII reference
1	Columbia River basalts	I	20±15	47° N	118° W	65° d	—	3°	66° d	11°	1°	+3°	87° N	40° E	48° N	38° N	1°	173°	$N+R$	Campbell & Runcorn (1956)
2	Siletz volcanics	*I*	60±10	45° N	124° W	63° d	57	70°	55° d	7°	−8°	+70°	37° N	50° W	36° N	48° N	35°	70°	$N+R$	Cox (1957)
3	Laney shales	*S*	60+10	42° N	109° W	61° d	30	353°	60° d	—	−1°	−7°	86° N	164° W	41° N	42° N	3°	270°	N	Torreson *et al.* (1949)
4	Dakota sst	*S*	100±20	36° N	113° W	56° d	23	344°	62° d	7°	6°	−16°	77° N	127° E	43° N	31° N	10°	228°	$N+R$	Runcorn (1956)
5	Springdale sst	*S*	170±20	36° N	113° W	56° d	55	338°	16° d	9°	−40°	−22°	55° N	107° E	8° N	9° N	43°	328°	N	du Bois *et al.* (1957)
6	Connecticut Valley lavas	*I*	170±20	42° N	73° W	61° d	—	10°	14° d	11°	−47°	+10°	55° N	89° E	7° N	6° N	48°	13°	$N+R$	du Bois *et al.* (1957)
7	Triassic (various) (Graham)	*S*	170±20	36° N	107° W	56° d	58	0°	45° d	—	−11°	0°	81° N	73° E	27° N	31° N	11°	0°	N	Graham (1955)
8	Brunswickian of Newark series	*S*	170±20	41° N	75° W	60° d	71	6°	28° d	3°	−32°	+6°	63° N	93° E	15° N	14° N	33°	10°	N	du Bois *et al.* (1957)
9	Supai beds (Graham)	*S*	200±20	36° N	113° W	56° d	59	330°	3° up	5°	−59°	−30°	41° N	117° E	1° S	0°	64°	326°	N	Graham (1955)
10	Supai beds (Runcorn)	*S*	200±20	36° N	113° W	56° d	56	313°	13° up	8°	−69°	−47°	26° N	121° E	7° S	11° S	88°	315°	N	Runcorn (1956)
11	Supai beds (Doell)	*S*	200±20	36° N	113° W	56° d	—	326°	8° up	7°	−64°	−34°	39° *N*	115° E	4° S	3° S	70°	324°	N	Doell (1955)
12	Naco sst.	*S*	250±30	36° N	113° W	56° d	32	330°	3° d	3°	−53°	−30°	41° N	120° E	1° N	1° N	57°	323°	N	Runcorn (1956)
13	Barnett formation	*S*	250±30	31° N	99° W	50° d	94	320°	2° up	—	−52°	−40°	42° N	135° E	1° S	8° N	63°	314°	N	Howell & Martinez (1957)
14	Clinton iron ore	*S*	330±20	34° N	87° W	53° d	—	321°	19° up	—	−72°	−39°	35° N	138° E	10° S	5° N	80°	323°	N	Howell *et al.* (1958)
15	Rosehill beds	*S*	330±20	40° N	78° W	59° d	—	322°	39° up	5°	−20°	−38°	19° N	138° E	22° S	7° S	76°	331°	R	Graham (1949)
16	Wilbern's fm. Point Peak member	*S*	450±50	31° N	99° W	50° d	180	129°	33° d	—	−17°	−51°	0°	158° E	18° S	5° S	85°	319°	N	Howell *et al.* (1958)

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Table 3. Australia

	I geological formation	II ig sed	III T	IV λ_0	IV L_0	V I_0	VI N	VII D	VII I	VIII δ	IX ΔI	X ψ	XI λ_p	XI L_p	XII λ	XIII λ_m	XIV ϕ	XV α	XVI N or R	XVII reference
1	Newer Volcanics, Victoria	I	7± 7	38° S	144° E	57° up	117	3°	60° up	4·8°	−3°	+3°	86° S	102° E	41° S	26° S	3°	30°	$N+R$	Irving & Green (1957)
2	Older Volcanics, Victoria	I	60±20	38° S	146° E	57° up	45	17°	73° up	6·8°	−16°	+17°	67° S	123° E	58° S	45° S	17°	17°	$N+R$	Irving & Green (1957)
3	dolerite sills, Tasmania	I	130±20	42° S	147° E	61° up	60	325°	85° up	3·5°	−24°	+35°	50° S	157° E	80° S	58° S	34°	5°	N	Irving (1956*b*)
4	Brisbane tuff	S	170±20	27° S	154° E	46° up	—	35°	83° up	—	−37°	+35°	39° S	143° E	76° S	72° S	50°	5°	$N+R$	Irving & Green (1958)
5	Volcanics, Illawarra	I	200±20	35° S	151° E	54° up	—	67°	81° d	11°	−27°	+67°	27° S	169° E	72° S	58° S	34°	15°	R	Irving & Green (1958)
6	Volcanics, Hunter Valley	I	200±20	33° S	151° E	52° up	—	110°	80° d	—	−28°	+110°	38° S	174° E	71° S	53° S	43°	14°	R	Irving & Green (1958)
7	Kuttung seds	S	250±30	33° S	151° E	52° up	75	270°	84° up	6°	−32°	+90°	32° S	165° E	78° S	61° S	39°	10°	$N+R$	Irving & Green (1958)
8	Kuttung lavas	I	250±30	33° S	151° E	52° up	—	5°	85° up	8°	−33°	+5°	43° S	150° E	80° S	66° S	33°	1°	N	Irving & Green (1958)
9	Ainslie volcanics	I	300±20	35° S	149° E	54° up	—	17°	30° up	12°	+24°	+17°	66° S	12° E	16° S	9° S	27°	146°	N	Irving & Green (1958)
10	Mugga porphyry	I	330±20	35° S	149° E	54° up	—	26°	30° up	22°	+24°	+26°	60° S	23° E	16° S	10° S	30°	131°	N	Irving & Green (1958)
11	Elder Mt. sandstone	S	450±50	16° S	126° E	30° up	—	231°	15° up	10°	+45°	+51°	34° S	8° E	8° S	13° N	68°	126°	R	Irving & Green (1958)
12	Antrim Plateau basalts	I	450±50	16° S	126° E	30° up	—	53°	2° up	12°	+28°	+53°	36° S	26° E	1° S	0°	58°	110°	$N+R$	Irving & Green (1958)

Table 4. India

	I geological formation	II ig sed	III T	IV λ_0	IV L_0	V I_0	VI N	VII D	VII I	VIII δ	IX ΔI	X ψ	XI λ_p	XI L_p	XII λ	XIII λ_m	XIV ϕ	XV α	XVI N or R	XVII reference
1	Rhyolites, above Deccan Traps Miocene?	I	35±10	22° N	72° E	39° d	15	355°	17° d	10°	−22°	−5°	75° N	89° W	9° N	7° N	22°	347°	N	Deutsch *et al.* (1958)
2	Upper Deccan Traps	I	60±10	17° N	74° E	31° d	197	341°	24° up	10°	−55°	−19°	54° N	75° W	13° S	11° S	59°	340°	N	Deutsch *et al.* (1958)
3	Lower Deccan Traps	I	100±20	18° N	74° E	33° d	—	149°	56° d	—	−89°	−31°	28° N	78° N	37° S	37° S	82°	343°	R	Irving (1956*a*)
4	Lower Deccan Traps	I	100±20	18° N	74° E	33° d	581	156°	56° d	10°	−89°	−24°	31° N	83° W	37° S	34° S	88°	347°	R	Clegg *et al.* (1956, 1957); Deutsch *et al.* (1958)
5	Rajmahal Traps	I	130±20	25° N	88° E	43° d	33	328°	64° up	6°	−107°	−32°	13° N	70° W	46° S	44° S	110°	346°	N	Clegg *et al.* (1958)

Table 5. Africa

	I geological formation	II ig sed	III T	IV λ_0	IV L_0	V I_0	VI N	VII D	VII I	VIII δ	IX ΔI	X ψ	XI λ_p	XI L_p	XII λ	XIII λ_m	XIV ϕ	XV α	XVI N or R	XVII reference
1	Karoo basalt	I	140±20	18° S	26° E	33° up	44	328°	40° up	4½°	−7°	−32°	63° N	78° W	23° S	5° S	26°	292°	N	Nairn (1956)
2	Karoo dolerite (1)	I	140±20	30° S	29° E	49° up	149	352°	62° d	12°	+111°	−8°	76° N	128° W	43° S	11° S	13°	343°	$N+R$	Graham & Hales (1957)
3	Karoo dolerite and baked sed. (2)	I	140±20	28° S	30° E	47° up	64	337°	59° d	<10°	+106°	−23°	76° N	128° W	40° S	11° S	18°	319°	$N+R$	Graham & Hales (1957)
4	Cave sandstone	S	170±20	23° S	27° E	40° up	44	326°	16° up	9½°	+24°	−34°	54° S	136° E	8° S	11° S	38°	241°	N	Creer *et al.* (1958)
5	Maji Ya Chumvi	S	200±20	3° S	39° E	6° up	21	267°	38° d	11°	+44°	+87°	40° N	150° E	21° S	35° S	84°	52°	N?	Creer *et al.* (1958)
6	Taru grit	S	200±20	3° S	39° E	6° up	32	87°	61° d	16½°	+67°	+87°	0°	87° E	42° S	28° N	85°	29°	N?	Creer *et al.* (1958)
7	Dwyka varved clays	S	250±30	18° S	29° E	33° up	10	360°	81° up	5½°	−48°	0°	36° S	29° E	72° S	55° N	48°	0°	$N+R$	Creer *et al.* (1958)
				18° S	29° E	—	9	333°	76° d	7°	+109°	−27°	7° N	17° E	64° S	80° N	67°	353°	$N+R$	Creer *et al.* (1958)

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Appendix III. Holmes *B* time scale

	age (My)	Period
Cenozoic (Tertiary)	50	Quaternary Pliocene Miocene Oligocene Eocene
Mesozoic	100	Cretaceous
	150	Jurassic Triassic
Palaeozoic	200	Permian
	250	Carbon-iferous
	300	Devonian or Old Red Sandstone
	350	Silurian
	400	Ordovician
	450	Cambrian
	500	
Pre-Cambrian		late Pre-Cambrian

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